

Porichoy (পরিচয়)

DPP-in-a-Box — every Bangladeshi garment gets a verifiable identity before the EU's 2027 Digital Product Passport wall.

rudra496.github.io/porichoy · Sylhet, Bangladesh · Rudra Sarker & team, SUST IPE

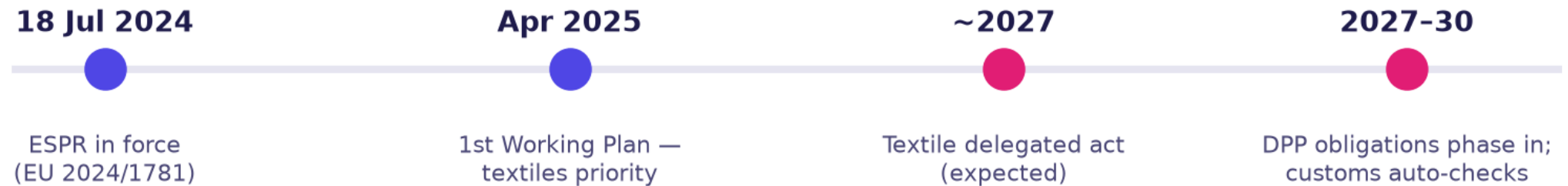
THE PROBLEM

EU customs will auto-check Digital Product Passports on imported garments: ESPR in force 18 Jul 2024 · textiles in the first Working Plan (Apr 2025)
• **textile delegated act expected ~2027. No passport data → no EU order.**

The data already exists in every factory — trapped in mixed Bangla/English Excels, ERP exports and paper trim cards. Global traceability platforms are brand-side and enterprise-priced; nobody captures at the Bangladeshi factory floor.

Most affected: small & mid-size exporters without compliance teams — and, through them, the ~4 million garment workers (BGMEA; majority women) whose livelihoods ride on Europe, the industry's biggest market.

The dated wall: EU Digital Product Passport timeline (European Commission)



OUR SOLUTION

Porichoy turns a factory’s existing records into buyer-ready EU Digital Product Passports in four steps — all live today, offline-tolerant, Bangla + English:

- 1 · INGEST — drop the factory Excel/CSV; every sheet parsed on-device. Nothing leaves the factory (local-first).
 - 2 · MAP — the engine reads chaotic Bangla/English headers with confidence scores; a trained steward confirms; the factory’s vocabulary is learned.
 - 3 · SCORE — DPP Readiness 0–100 per production order (weights = ESPR Art. 8 categories) with the exact gap list.
 - 4 · PASSPORT — one click → a QR whose link embeds the passport; scanning opens the buyer view on any phone. No backend.
- 19/19 automated engine tests · SHA-256 provenance chain · stewards fix gaps in-place and the score updates instantly.

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Dashboard Evidence Method & sources বাংলা

DPP Readiness

Weighted share of EU required fields you already evidence

65

Compliance gaps

Shonal Textiles Ltd (sample data) · 4 production orders

Bring records in

Upload a real factory Excel — or try the bundled messy sample first.

Upload records Load sample data

Download readiness report (CSV)

Local-first: in this pilot your raw records never leave the factory device. Only the passport you choose to share is public.

Provenance chain live — 4 11kbs · 13p c4bd7424ba78e...

Production orders

PO	PRODUCT	DPP READINESS	MISSING	
PO-1001	Mens Polo Shirt S/M	90	glin, hs_code	Open
PO-1002	Ladies Denim Jacket	90	glin, hs_code	Open
PO-1003	Kids T-Shirt	45	glin, hs_code, recycled_pct, facilities, certs, recycling	Open
PO-1004	—	35	glin, hs_code, product_name, fiber_comp, recycled_pct, mfg_dates, care_info, recycling, substances	Open

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PO-1002 — Ladies Denim Jacket

90

Buyer-ready

90/100 weight-points evidenced and validated.

Publish for the buyer

The passport snapshot rides inside the QR itself — opens on any phone, no cloud needed.

Open buyer view

Copy link

Download passport (JSON)

Required attribute checklist

ATTRIBUTE	VALUE	SOURCE COLUMN	REGULATORY ANCHOR	WEIGHT
Production order / lot ID	PO-1002	vID ref	ESPR Art.8(a) unique identifier	10
Product code (GTIN/SKU)	missing Add	—	ESPR Art.8(a) + GS1	5
HS tariff code	missing Add	—	Customs linkage	5
Product description	Ladies Denim Jacket	Item Description	ESPR Art.8(b)	5

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Evidence base

Every paper below was retrieved from the Crossref API and its DOI re-resolved live on 2026-09-13. Titles, journals, years and authors are copied verbatim from the API response (never typed by hand). The base grounds Porichoy's problem statement (EU regulation, traceability, Bangladesh RMG compliance) and its design choices (circularity, data provenance).

DPP & regulation · 5 traceability technology · 2 Bangladesh RMG compliance · 1 Bangladesh circularity · 1 circular economy · 2 sustainable supply chain · 1 due diligence & buyers · 1 circularity & recycling · 1 textile waste impact · 1 DPP & tracking · 1

#	PAPER	VENUE	YEAR	GROUP	CITED BY	DOI (VERIFIED)
1	The Circular Economy – A new sustainability paradigm?	Journal of Cleaner Production	2017	circular economy	7,922	10.1016/j.jclepro.2016.12.008
2	Conceptualizing the circular economy: An analysis of 114 definitions	Resources, Conservation and Recycling	2017	circular economy	7,289	10.1016/j.resconres.2017.09.005
3	From a literature review to a conceptual framework for sustainable supply chain management	Journal of Cleaner Production	2008	sustainable supply chain	4,930	10.1016/j.jclepro.2008.04.008
4	Blockchain-Enabled Supply Chain Traceability in the Textile and Apparel Supply Chain: A Case Study of the Fiber Producer, Lenzing	Sustainability	2021	traceability technology	94	10.3390/su131318095
5	Rebeyond the Circular Economy in Materials	Journal of Cleaner Production	2024	circularity & recycling	4	10.1016/j.jclepro.2024.140100

Live production app — screenshots fetched 2026-09-13 · rudra496.github.io/porichoy



MARKET SIZE & OPPORTUNITY

A dated compliance wall creates a forced-adoption market:

≈ 4 million RMG workers (BGMEA) · world's #2 ready-made garment exporter · Europe is the biggest market (NIC 3.0 call) · thousands of export factories in the BGMEA member base — every one shipping to the EU must be DPP-ready inside the 2027–30 window.

Illustrative revenue math on Porichoy's own pricing (BDT 2,000–8,000/factory/month):

100 pilot-scale factories ≈ BDT 60 lakh/year · 1,000 factories ≈ BDT 6 crore/year — before brand-funded supplier-onboarding contracts, where buyers pay to bring their supplier base onto passports before the deadline.

≈4 million

RMG workers, majority women (BGMEA)

#2 exporter

world RMG — Europe the biggest market

2027–30

EU DPP phase-in; customs auto-checks

10→100+

factories: year-1 pilot → year-3 scale

Timing is the wedge: buyers already ask, budgets activate before the deadline, and no Bangladesh-built tool serves the factory floor today — the verified competitors are brand-side only.

BUSINESS & REVENUE MODEL

Three simple revenue lines:

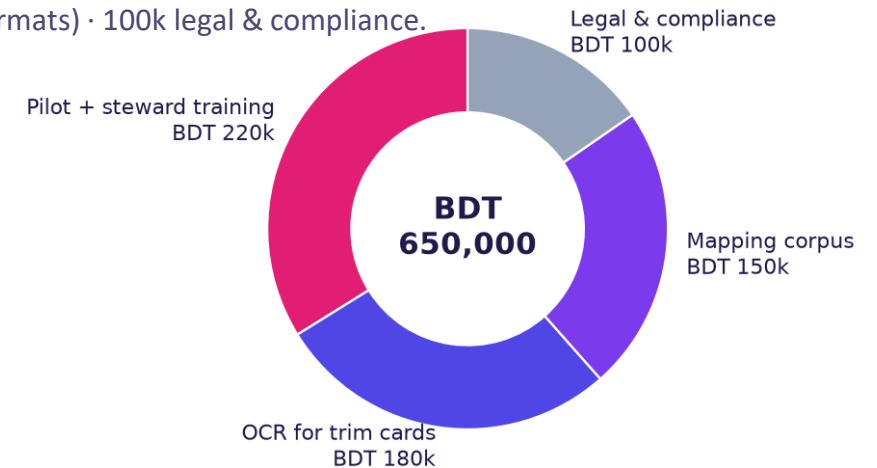
1. Factory subscriptions — BDT 2,000–8,000/month by size (SME-priced, ~10× below enterprise platforms).
2. Onboarding & data-rescue — one-time fee: steward training + first ingest + mapping historic files.
3. Brand-funded supplier onboarding — buyers pay to make their supplier base passport-ready before 2027–30.

Unit economics: onboarding ≈ 2 person-days/factory · software gross margin >80% · replication is a template, not a project.

Pollination grant (BDT 650,000) funds the 10-factory pilot:

BDT 220k pilot + steward training · 180k OCR for paper trim cards · 150k mapping corpus (Bangla + ERP formats) · 100k legal & compliance.

Pollination grant allocation

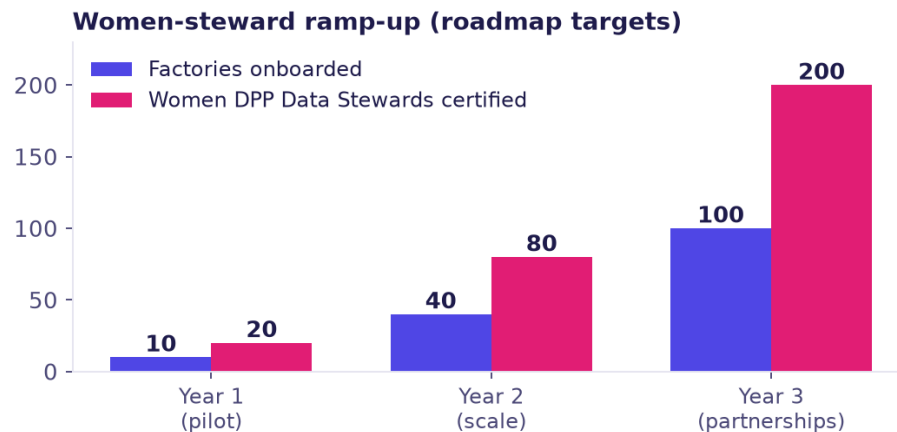


OUR IMPACT

Competitiveness: EU-order eligibility is protected (customs auto-checks), compliance prep per PO drops from days of file-hunting to under an hour, and readiness scores make factories the buyer-preferred choice.

Women: two women operators per factory are trained & certified as DPP Data Stewards — formal, higher-skilled roles (20 in year 1 → 200 by year 3, targets) — and keeping factories order-eligible protects a majority-women workforce (≈4M workers, BGMEA).

Circularity: DPP fields make recycled content, certificates and end-of-life routes verifiable — turning circularity claims into machine-checkable data that flows to buyers and recyclers.



Left: women-steward ramp (roadmap targets). Right: RMG factory floor — photo CC BY 2.0, Wikimedia Commons.

OUR TEAM



Rudra Sarker — Founder & full-time product engineer. Final-year B.Sc. Industrial & Production Engineering, SUST. Builds every part of Porichoy: the EU-aligned DPP schema, the Bangla/English mapping engine, readiness scoring and the provenance chain.

Awards & achievements (verified): Q1 journal publication — Journal of Ethnopharmacology (Elsevier, 2026), 3rd author · Z.AI Global Ambassador (since Nov 2025) · IEF Bangladesh Ambassador (since Apr 2026) · Secretary, RoboSUST (2025–26) · GaoTek industrial internship (2024–25).

Two IPE backgrounds ground us in how factories actually run; systems engineering makes the software ship. Women lead our steward program by design — the same people the product trains and employs. No big-name advisers: 19/19 passing tests, public code and a live product speak instead.

Lutfay Hodayra Islam

SUST · IPE

Factory Operations & Pilot Lead

Process mapping, factory data collection, readiness audits across the pilot factories

Priom Sarker

BUET

Systems & Technology Lead

Systems architecture, ERP/API integrations, technical review of the ingestion pipeline

Shuvo Kundu

University of Asia Pacific

Data & Research Lead

Dataset curation, Bangla mapping-corpus growth, literature & research support

Samia Hossain

Steward Training & Outreach Lead

Runs the women DPP Data Steward training program and factory outreach

Live: rudra496.github.io/porichoy · code: github.com/rudra496/porichoy · 19/19 engine tests · launched Sep 2026

Everything we claim is checkable — that is the product's whole point.

THANK YOU

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Live product: rudra496.github.io/porichoy · Code: github.com/rudra496/porichoy